Data Preparation

1. Demographic and pre-processed physiological data files were downloaded from the Brain Body Behaviour Dataset website: https://bbbd.pythonanywhere.com
2. MAT files were converted to CSV using Python scripts written with the assistance of Claude
    a. sample script: https://github.com/joanna-wi/joanna-wi.github.io/blob/Project-2/Scripts%20-%20Project%202/mat2csv_conversion.py
3. Based on original EEG and Heart Rate data, additional metrics were calculated and converted into csv files using Python scripts written with the assistance of Claude:
    a. absolute PSD: https://github.com/joanna-wi/joanna-wi.github.io/blob/Project-2/Scripts%20-%20Project%202/eeg_psd_absolute_power.py
    b. FAA: https://github.com/joanna-wi/joanna-wi.github.io/blob/Project-2/Scripts%20-%20Project%202/eeg_faa_calculation.py
    c. RMSSD: https://github.com/joanna-wi/joanna-wi.github.io/blob/Project-2/Scripts%20-%20Project%202/calculate_rmssd.py
4. CSV files too large to upload to PowerBI directly (HeartRate and PSD files) were converted to PARQUET using SQL in DuckDB.
5. Following participant were excluded from further analysis:
    a. 30, 31,32 – due to missing physiological data
    b. 8, 14, 20 – due to missing demographic data
6. Column names, data types and formatting were adjusted; unnecessary columns were removed.
7. In sources containing physiological data, outlier removal was conducted using the Interquartile Range Method in M code (values outside of Q1 - 1.5 * IQR and Q3 + 1.5 * IQR were filtered out).
8. Helper columns for participant segmentation were created:
    a. "Age range" – dividing participants into two segments based on mean sample age of 26
    b. "Occupation type" – dividing participants into two segments based on most common value in the occupation field (student status)
    c. "ASRS/Digit Span score" – dividing participants into two segments based on sample mean score
9. Where relevant, columns with participant-level mean-centred data were constructed
10. Additional tables were constructed for:
    a. Assignment of stimuli to video categories (PA, WA, PG)
    b. Preparation of data for ASRS/Digit Span histogram visual
    c. ASRS/Digit Span mean display card visual
    d. Condition classification (Attentive/Distracted)
    e. Feedback attribute classification (Educational/Engaging/Enjoyable)
    f. Creation of participant-stimulus keys
    g. Representation of mean PSD values per frequency band per site for the heatmap visual
11. Statistics – correlations between behavioural and physiological data
    a. Spearman correlations were inspected in Jamovi and significant results reported ("Correlation_" files): https://github.com/joanna-wi/joanna-wi.github.io/tree/Project-2/Statistics%20-%20Project%202
12. Segmentation
    a. Cross-scale Spearman correlations of raw and mean-centred feedback scores were inspected in Jamovi to determine the optimal approach to hierarchical clustering. Due to high correlation between Engagement and Enjoyment attributes (results in "Segmentation_Correlations_" files: https://github.com/joanna-wi/joanna-wi.github.io/tree/Project-2/Statistics%20-%20Project%202), the two were subsequently collapsed into one averaged score.
    b. Two separate hierarchical clustering analyses were conducted in Jamovi, one based on Educational dimension scores, the other based on Engagement-Enjoyment composite scores. A

minimum cluster size of 5 respondents was used as a criterion when choosing the number of clusters (results in "Clustering_" files: https://github.com/joanna-wi/joanna-wi.github.io/tree/Project-2/Statistics%20-%20Project%202).

c. Cluster assignment from each of the two analyses was mapped in Excel onto the "participants" file. These were subsequently cross-combined into four groups (A–D), covering every pairing of cluster membership.
d. Demographic profile and feedback score pattern of each segment were inspected in Excel.
e. Physiological data were not incorporated into the analysis due to small cluster sizes and resultant sensitivity to intersubject variability.

Dashboard Construction

1. Measures were created to obtain:
    a. Mean scores and physiological metrics
    b. SEM values for error bars
    c. Dynamic titles
    d. "Selected metric" variables for visuals with slicers
2. Calculated tables were created to obtain:
    a. Slicers
    b. Length of each video, for visuals representing temporal changes in physiological metrics
    c. Mean physiological metrics per each 5 second bin of video exposure, for visuals representing temporal changes in physiological metrics
3. Buttons, bookmarks, alongside panels for insights and filters were implemented for report navigation.

Report Preparation

1. The report was built in PowerPoint.
2. The bar graph miniatures were constructed in Excel using data from the dashboard.
3. Segmentation graphs were built in Excel based on hierarchical clustering results.
4. Image from the title slide was downloaded from Pexels: https://www.pexels.com/photo/person-holding-space-gray-iphone-5-34407/
5. Some icons were sourced from the PowerPoint library, whereas others were downloaded from Flaticon:
    a. Heart
    b. Brain
    c. Eye
    d. Background
    e. Objectives
    f. Survey
    g. Study setup
    h. Online learning
    i. Behavioural test
    j. Respondents
    k. Motion
    l. Brain in hands
    m. Segmentation